delhivery

July 25, 2025

0.0.1 Business Problem:

Delhivery is the largest and fastest-growing fully integrated player in India by revenue in Fiscal 2021. They aim to build the operating system for commerce, through a combination of world-class infrastructure, logistics operations of the highest quality, and cutting-edge engineering and technology capabilities. The Data team builds intelligence and capabilities using this data that helps them to widen the gap between the quality, efficiency, and profitability of their business versus their competitors.

The company wants to understand and process the data coming out of data engineering pipelines:

- Clean, sanitize and manipulate data to get useful features out of raw fields
- Make sense out of the raw data and help the data science team to build forecasting models on it

[57]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
import re
from scipy.stats import ttest_rel
```

[58]:
```
df_dhy = pd.read_csv('C:
↪\\Users\\njain121\\Desktop\\Scalar\\CaseStudy\\Delhivery\\delhivery_data.
↪csv')
df_dhy.head()
```

[58]:
```
       data          trip_creation_time  \
0  training  2018-09-20 02:35:36.476840
1  training  2018-09-20 02:35:36.476840
2  training  2018-09-20 02:35:36.476840
3  training  2018-09-20 02:35:36.476840
4  training  2018-09-20 02:35:36.476840

                                 route_schedule_uuid route_type  \
0  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
1  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
2  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
```

```
3  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting
4  thanos::sroute:eb7bfc78-b351-4c0e-a951-fa3d5c3…    Carting

                trip_uuid source_center                 source_name  \
0  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
1  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
2  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
3  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)
4  trip-153741093647649320  IND388121AAA  Anand_VUNagar_DC (Gujarat)

  destination_center               destination_name  \
0       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)
1       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)
2       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)
3       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)
4       IND388620AAB  Khambhat_MotvdDPP_D (Gujarat)

               od_start_time  …            cutoff_timestamp  \
0  2018-09-20 03:21:32.418600  …         2018-09-20 04:27:55
1  2018-09-20 03:21:32.418600  …         2018-09-20 04:17:55
2  2018-09-20 03:21:32.418600  …  2018-09-20 04:01:19.505586
3  2018-09-20 03:21:32.418600  …         2018-09-20 03:39:57
4  2018-09-20 03:21:32.418600  …         2018-09-20 03:33:55

   actual_distance_to_destination  actual_time  osrm_time osrm_distance  \
0                       10.435660         14.0       11.0       11.9653
1                       18.936842         24.0       20.0       21.7243
2                       27.637279         40.0       28.0       32.5395
3                       36.118028         62.0       40.0       45.5620
4                       39.386040         68.0       44.0       54.2181

     factor  segment_actual_time  segment_osrm_time  segment_osrm_distance  \
0  1.272727                 14.0               11.0                11.9653
1  1.200000                 10.0                9.0                 9.7590
2  1.428571                 16.0                7.0                10.8152
3  1.550000                 21.0               12.0                13.0224
4  1.545455                  6.0                5.0                 3.9153

   segment_factor
0        1.272727
1        1.111111
2        2.285714
3        1.750000
4        1.200000

[5 rows x 24 columns]
```

0.0.2 Column Profiling:

- data - tells whether the data is testing or training data
- trip_creation_time – Timestamp of trip creation
- route_schedule_uuid – Unique Id for a particular route schedule
- route_type – Transportation type
- FTL – Full Truck Load: FTL shipments get to the destination sooner, as the truck is making no other pickups or drop-offs along the way
- Carting: Handling system consisting of small vehicles (carts)
- trip_uuid - Unique ID given to a particular trip (A trip may include different source and destination centers)
- source_center - Source ID of trip origin
- source_name - Source Name of trip origin
- destination_cente – Destination ID
- destination_name – Destination Name
- od_start_time – Trip start time
- od_end_time – Trip end time
- start_scan_to_end_scan – Time taken to deliver from source to destination
- is_cutoff – Unknown field
- cutoff_factor – Unknown field
- cutoff_timestamp – Unknown field
- actual_distance_to_destination – Distance in Kms between source and destination warehouse
- actual_time – Actual time taken to complete the delivery (Cumulative)
- osrm_time – An open-source routing engine time calculator which computes the shortest path between points in a given map (Includes usual traffic, distance through major and minor roads) and gives the time (Cumulative)
- osrm_distance – An open-source routing engine which computes the shortest path between points in a given map (Includes usual traffic, distance through major and minor roads) (Cumulative)
- factor – Unknown field
- segment_actual_time – This is a segment time. Time taken by the subset of the package delivery
- segment_osrm_time – This is the OSRM segment time. Time taken by the subset of the package delivery
- segment_osrm_distance – This is the OSRM distance. Distance covered by subset of the package delivery
- segment_factor – Unknown field

[59]:
```
df_dhy.shape
```

[59]:
```
(144867, 24)
```

[60]:
```
df_dhy.info()  ## details of the dataframe
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144867 entries, 0 to 144866
Data columns (total 24 columns):
```

```
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   data                            144867 non-null  object
 1   trip_creation_time              144867 non-null  object
 2   route_schedule_uuid             144867 non-null  object
 3   route_type                      144867 non-null  object
 4   trip_uuid                       144867 non-null  object
 5   source_center                   144867 non-null  object
 6   source_name                     144574 non-null  object
 7   destination_center              144867 non-null  object
 8   destination_name                144606 non-null  object
 9   od_start_time                   144867 non-null  object
 10  od_end_time                     144867 non-null  object
 11  start_scan_to_end_scan          144867 non-null  float64
 12  is_cutoff                       144867 non-null  bool
 13  cutoff_factor                   144867 non-null  int64
 14  cutoff_timestamp                144867 non-null  object
 15  actual_distance_to_destination  144867 non-null  float64
 16  actual_time                     144867 non-null  float64
 17  osrm_time                       144867 non-null  float64
 18  osrm_distance                   144867 non-null  float64
 19  factor                          144867 non-null  float64
 20  segment_actual_time             144867 non-null  float64
 21  segment_osrm_time               144867 non-null  float64
 22  segment_osrm_distance           144867 non-null  float64
 23  segment_factor                  144867 non-null  float64
dtypes: bool(1), float64(10), int64(1), object(12)
memory usage: 25.6+ MB
```

[61]:
```
df_dhy.describe(include='all')
```

[61]:
```
             data          trip_creation_time  \
count      144867                      144867
unique          2                       14817
top      training  2018-10-01 05:04:55.268931
freq       104858                         101
mean          NaN                         NaN
std           NaN                         NaN
min           NaN                         NaN
25%           NaN                         NaN
50%           NaN                         NaN
75%           NaN                         NaN
max           NaN                         NaN

                                       route_schedule_uuid route_type  \
count                                               144867     144867
unique                                                1504          2
```

```
top     thanos::sroute:4029a8a2-6c74-4b7e-a6d8-f9e069f…        FTL
freq                                                 1812      99660
mean                                                  NaN        NaN
std                                                   NaN        NaN
min                                                   NaN        NaN
25%                                                   NaN        NaN
50%                                                   NaN        NaN
75%                                                   NaN        NaN
max                                                   NaN        NaN

                     trip_uuid source_center                    source_name  \
count                   144867        144867                         144574
unique                   14817          1508                           1498
top     trip-153837029526866991  IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)
freq                       101         23347                          23347
mean                       NaN           NaN                            NaN
std                        NaN           NaN                            NaN
min                        NaN           NaN                            NaN
25%                        NaN           NaN                            NaN
50%                        NaN           NaN                            NaN
75%                        NaN           NaN                            NaN
max                        NaN           NaN                            NaN

       destination_center               destination_name  \
count              144867                         144606
unique               1481                           1468
top          IND000000ACB  Gurgaon_Bilaspur_HB (Haryana)
freq                15192                          15192
mean                  NaN                            NaN
std                   NaN                            NaN
min                   NaN                            NaN
25%                   NaN                            NaN
50%                   NaN                            NaN
75%                   NaN                            NaN
max                   NaN                            NaN

                    od_start_time  …     cutoff_timestamp  \
count                      144867  …               144867
unique                      26369  …                93180
top     2018-09-21 18:37:09.322207  …  2018-09-24 05:19:20
freq                           81  …                   40
mean                          NaN  …                  NaN
std                           NaN  …                  NaN
min                           NaN  …                  NaN
25%                           NaN  …                  NaN
50%                           NaN  …                  NaN
75%                           NaN  …                  NaN
```

```
max                            NaN  …                  NaN

        actual_distance_to_destination    actual_time      osrm_time  \
count                     144867.000000  144867.000000  144867.000000
unique                              NaN            NaN            NaN
top                                 NaN            NaN            NaN
freq                                NaN            NaN            NaN
mean                         234.073372     416.927527     213.868272
std                          344.990009     598.103621     308.011085
min                            9.000045       9.000000       6.000000
25%                           23.355874      51.000000      27.000000
50%                           66.126571     132.000000      64.000000
75%                          286.708875     513.000000     257.000000
max                         1927.447705    4532.000000    1686.000000

        osrm_distance         factor  segment_actual_time  segment_osrm_time  \
count   144867.000000  144867.000000        144867.000000      144867.000000
unique            NaN            NaN                  NaN                NaN
top               NaN            NaN                  NaN                NaN
freq              NaN            NaN                  NaN                NaN
mean       284.771297       2.120107            36.196111          18.507548
std        421.119294       1.715421            53.571158          14.775960
min          9.008200       0.144000          -244.000000           0.000000
25%         29.914700       1.604264            20.000000          11.000000
50%         78.525800       1.857143            29.000000          17.000000
75%        343.193250       2.213483            40.000000          22.000000
max       2326.199100      77.387097          3051.000000        1611.000000

        segment_osrm_distance  segment_factor
count            144867.00000   144867.000000
unique                    NaN             NaN
top                       NaN             NaN
freq                      NaN             NaN
mean                 22.82902        2.218368
std                  17.86066        4.847530
min                   0.00000      -23.444444
25%                  12.07010        1.347826
50%                  23.51300        1.684211
75%                  27.81325        2.250000
max                2191.40370      574.250000

[11 rows x 24 columns]
```

[62]:
```python
print(df_dhy.isna().any().any())##Check for the missing values
print()
df_dhy.isna().sum()
```

```
True
```

[62]:
```
data                                0
trip_creation_time                  0
route_schedule_uuid                 0
route_type                          0
trip_uuid                           0
source_center                       0
source_name                       293
destination_center                  0
destination_name                  261
od_start_time                       0
od_end_time                         0
start_scan_to_end_scan              0
is_cutoff                           0
cutoff_factor                       0
cutoff_timestamp                    0
actual_distance_to_destination      0
actual_time                         0
osrm_time                           0
osrm_distance                       0
factor                              0
segment_actual_time                 0
segment_osrm_time                   0
segment_osrm_distance               0
segment_factor                      0
dtype: int64
```

[63]:
```
df_dhy['source_name'].fillna('NA', inplace=True)
df_dhy['destination_name'].fillna('NA', inplace=True)

warnings.filterwarnings("ignore")
```

[64]:
```
## Unique Attributes
df_dhy.nunique().sort_values(ascending=False)
```

[64]:
```
actual_distance_to_destination    144515
osrm_distance                     138046
segment_osrm_distance             113799
cutoff_timestamp                   93180
factor                             45641
od_start_time                      26369
od_end_time                        26369
trip_creation_time                 14817
trip_uuid                          14817
segment_factor                      5675
```

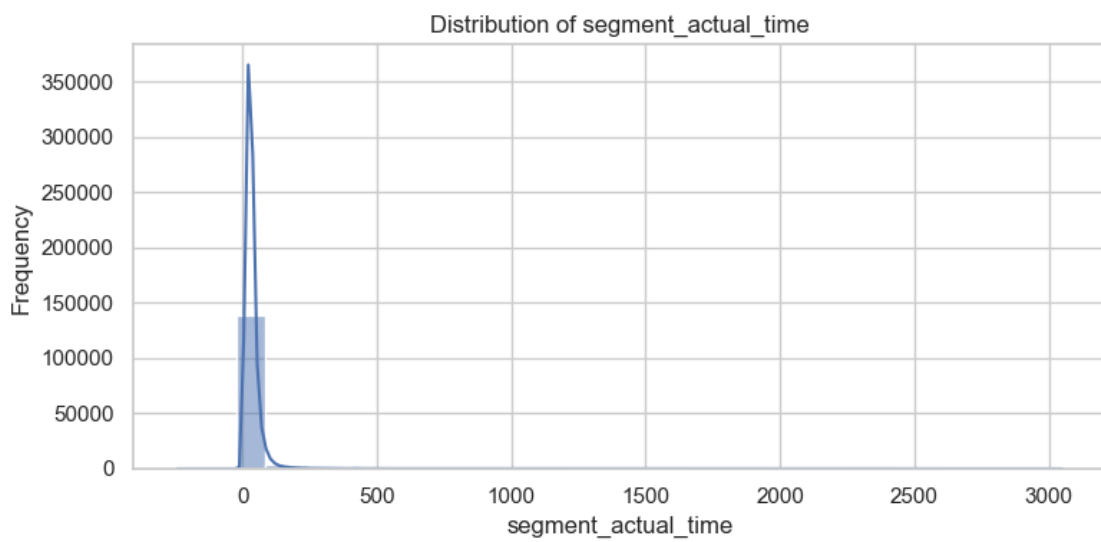
```
actual_time                          3182
start_scan_to_end_scan               1915
osrm_time                            1531
source_center                        1508
route_schedule_uuid                  1504
source_name                          1499
destination_center                   1481
destination_name                     1469
segment_actual_time                   747
cutoff_factor                         501
segment_osrm_time                     214
route_type                              2
data                                    2
is_cutoff                               2
dtype: int64
```

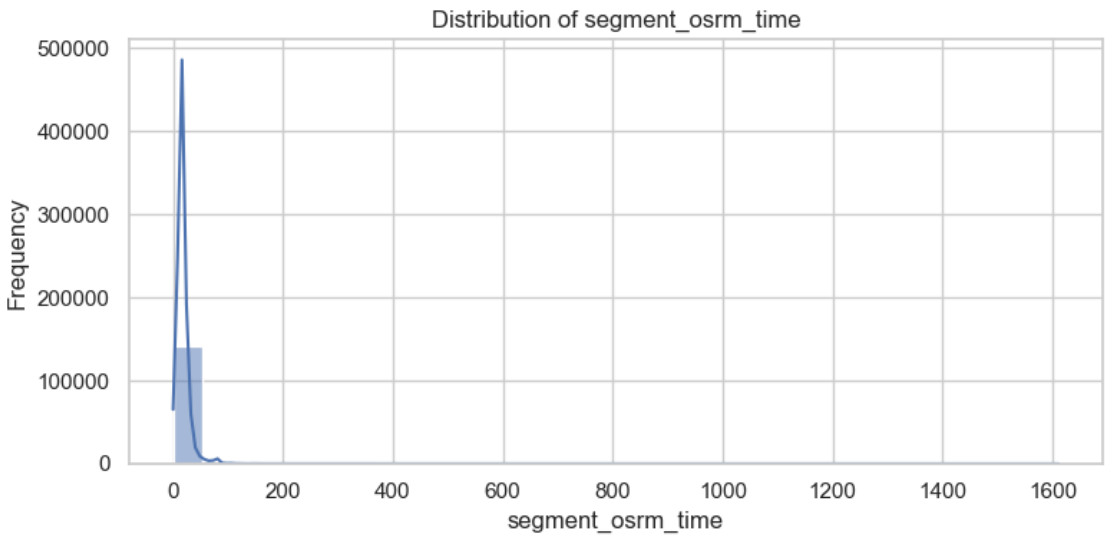
[65]:
```
## Identify if there is any duplicate row in the dataframe
duplicate_rows = df_dhy[df_dhy.duplicated()]
print("Duplicate rows identified:")
print(duplicate_rows)
```

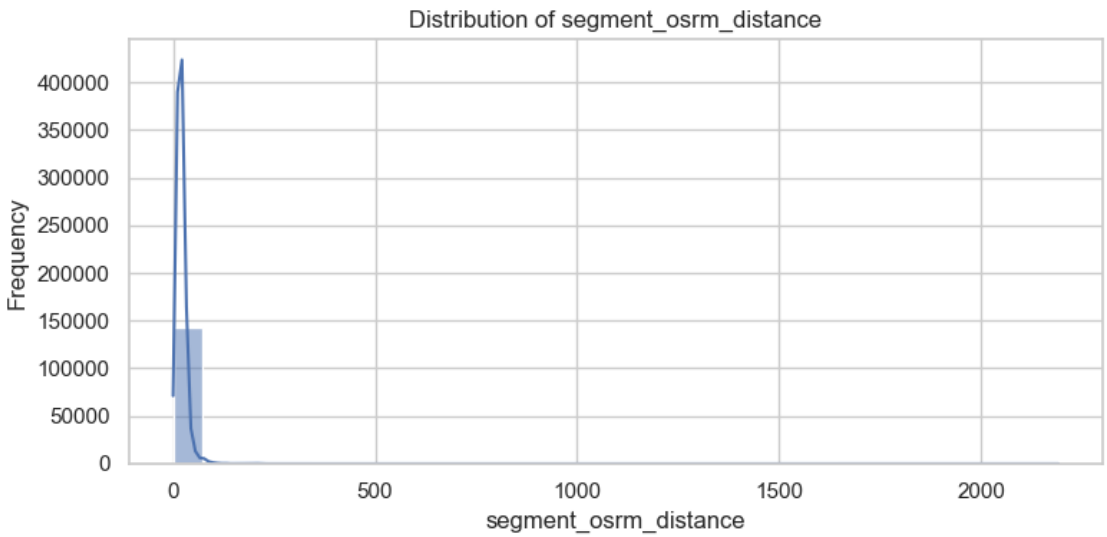
```
Duplicate rows identified:
Empty DataFrame
Columns: [data, trip_creation_time, route_schedule_uuid, route_type, trip_uuid,
source_center, source_name, destination_center, destination_name, od_start_time,
od_end_time, start_scan_to_end_scan, is_cutoff, cutoff_factor, cutoff_timestamp,
actual_distance_to_destination, actual_time, osrm_time, osrm_distance, factor,
segment_actual_time, segment_osrm_time, segment_osrm_distance, segment_factor]
Index: []

[0 rows x 24 columns]
```

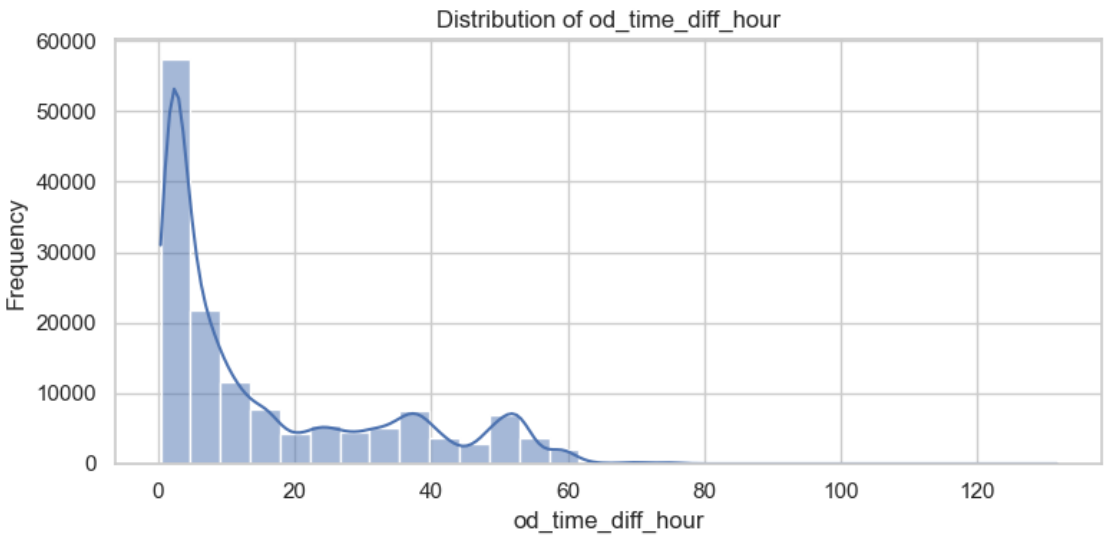
[66]:
```
# Convert datetime columns
df_dhy['od_start_time'] = pd.to_datetime(df_dhy['od_start_time'],␣
 ↪errors='coerce')
df_dhy['od_end_time'] = pd.to_datetime(df_dhy['od_end_time'], errors='coerce')

# Calculate od_time_diff_hour
df_dhy['od_time_diff_hour'] = (df_dhy['od_end_time'] - df_dhy['od_start_time']).
 ↪dt.total_seconds() / 3600

# Identify continuous and categorical variables
continuous_vars = ['segment_actual_time', 'segment_osrm_time',␣
 ↪'segment_osrm_distance', 'od_time_diff_hour']
categorical_vars = ['route_type', 'is_cutoff']

# Set up the plotting style
```

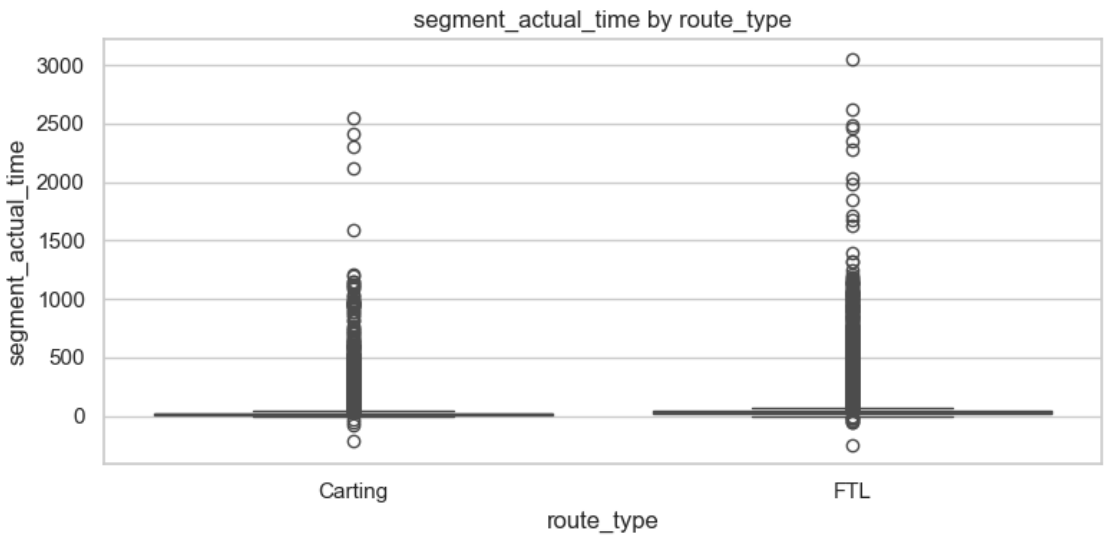
```python
sns.set(style="whitegrid")

# Create distribution plots for continuous variables
for col in continuous_vars:
    plt.figure(figsize=(8, 4))
    sns.histplot(df_dhy[col].dropna(), kde=True, bins=30)
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Frequency')
    plt.tight_layout()
    plt.show()

# Create boxplots for categorical variables against a continuous variable
for cat in categorical_vars:
    for cont in continuous_vars:
        plt.figure(figsize=(8, 4))
        sns.boxplot(x=cat, y=cont, data=df_dhy)
        plt.title(f'{cont} by {cat}')
        plt.xlabel(cat)
        plt.ylabel(cont)
        plt.tight_layout()
        plt.show()
```

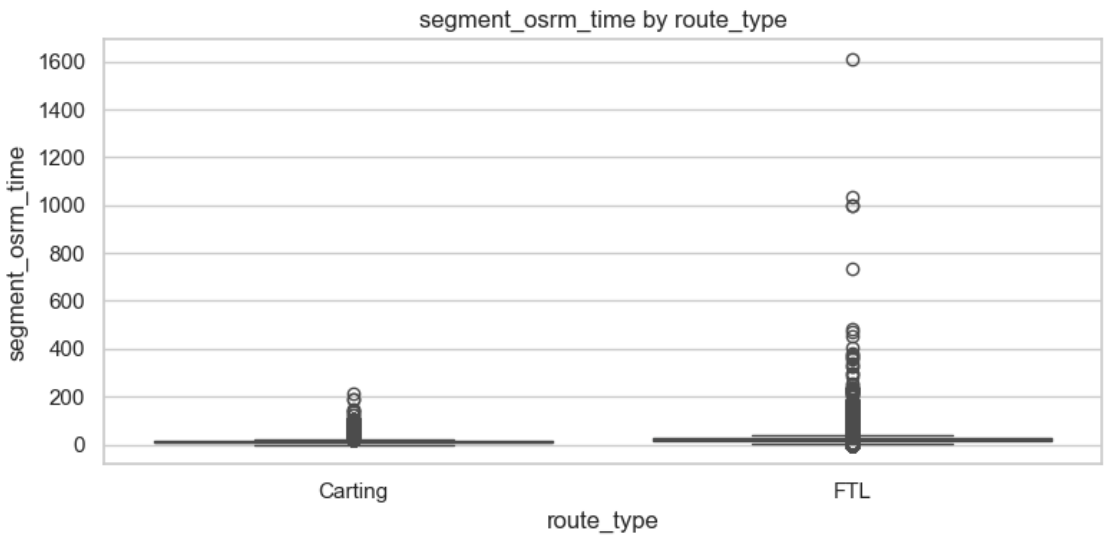


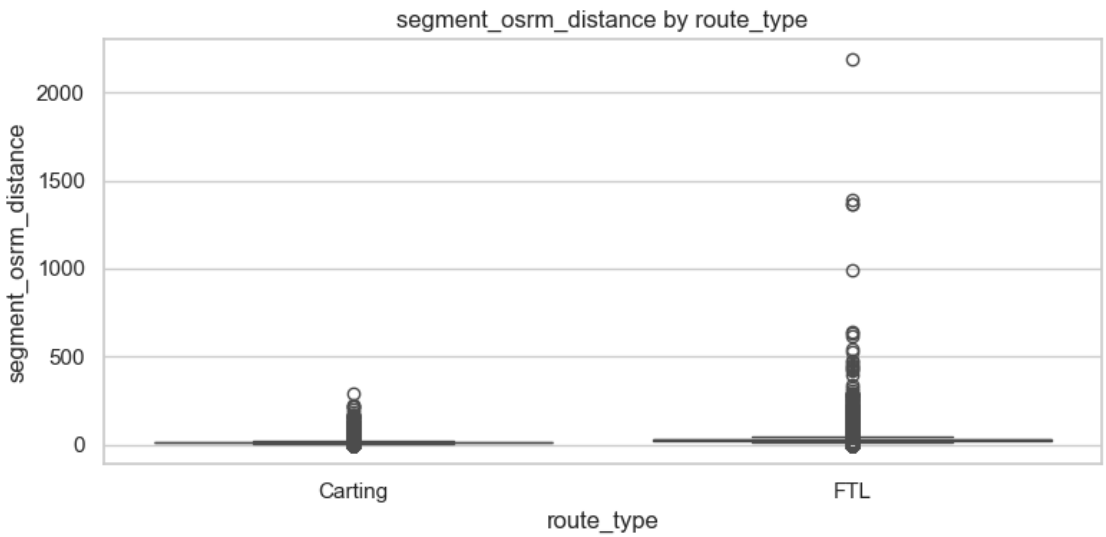

Distribution of segment_osrm_time
Frequency
500000
400000
300000
200000
100000
0
0
200
400
600
800
1000
1200
1400
1600
segment_osrm_time

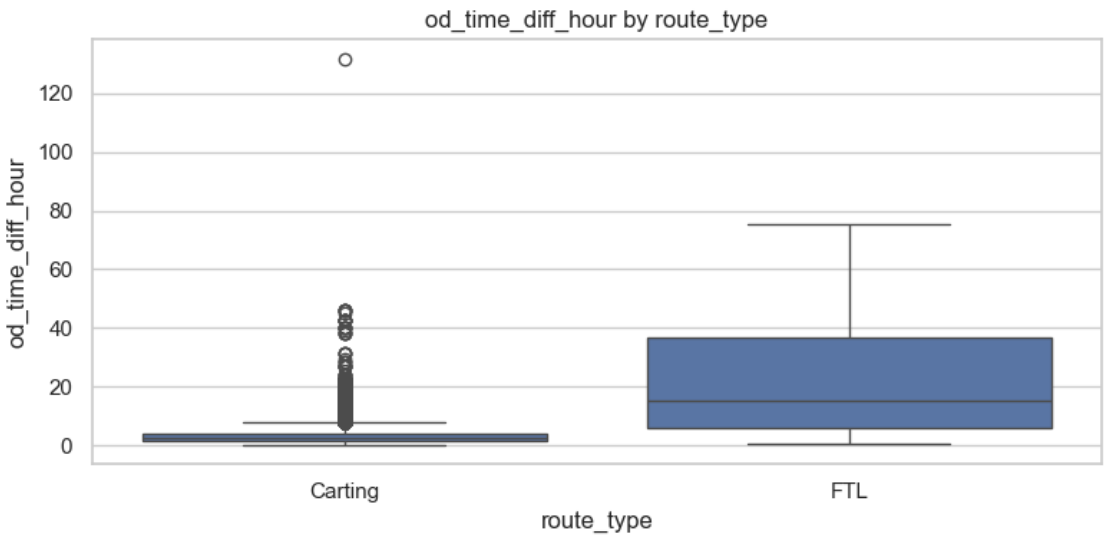

Distribution of segment_osrm_distance
Frequency
400000
350000
300000
250000
200000
150000
100000
50000
0
0
500
1000
1500
2000
segment_osrm_distance

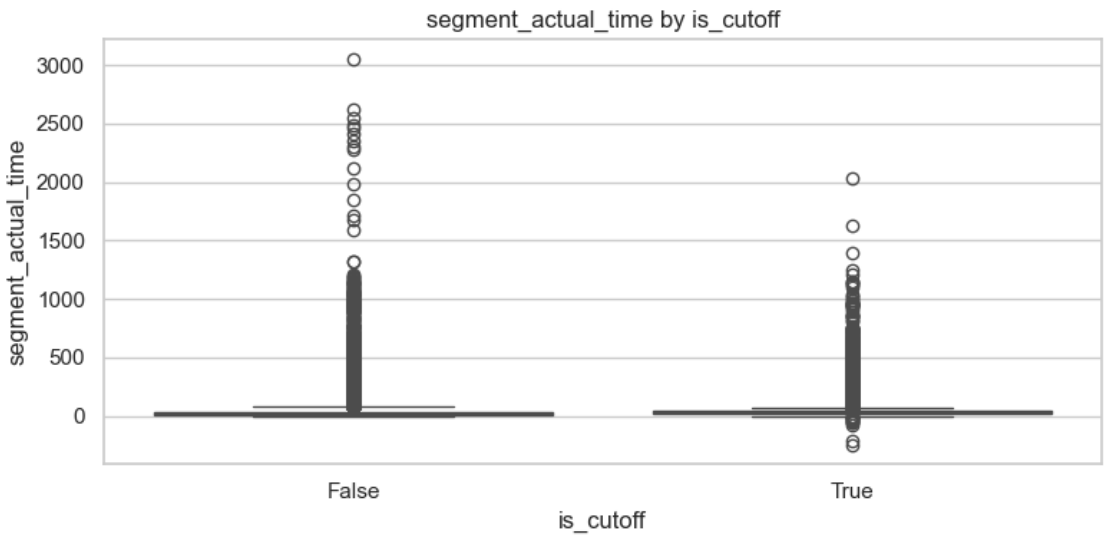

Distribution of od_time_diff_hour
Frequency
60000
50000
40000
30000
20000
10000
0
0
20
40
60
80
100
120
od_time_diff_hour

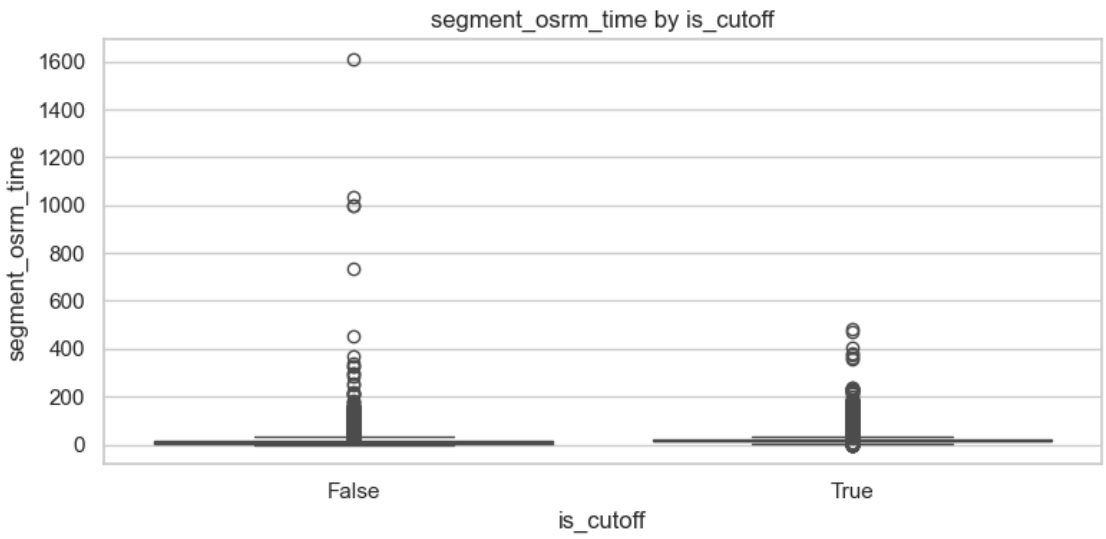

segment_actual_time by route_type
segment_actual_time
3000
2500
2000
1500
1000
500
0
Carting
FTL
route_type

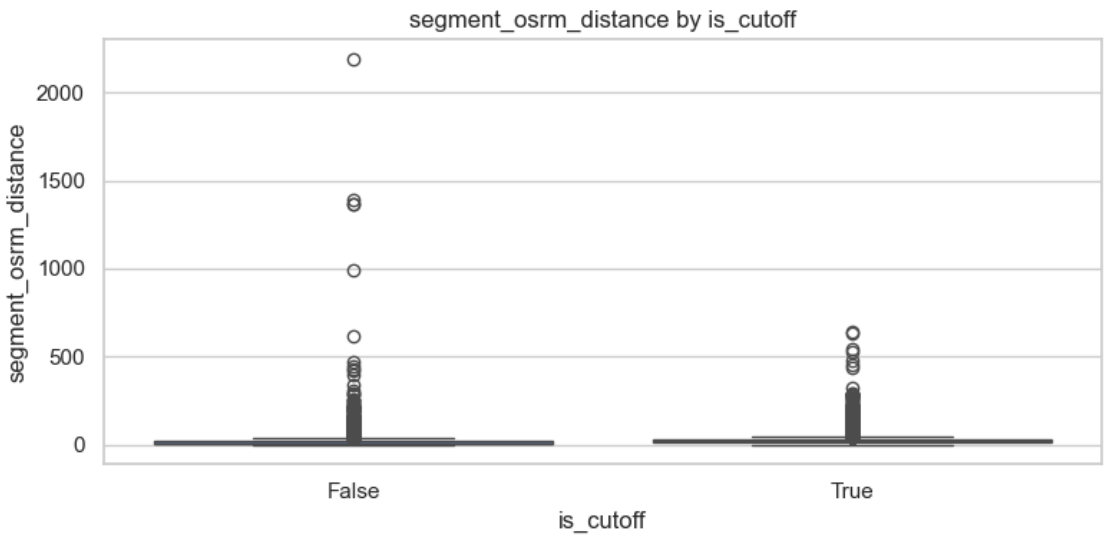

segment_osrm_time by route_type
segment_osrm_time
1600
1400
1200
1000
800
600
400
200
0
Carting
FTL
route_type

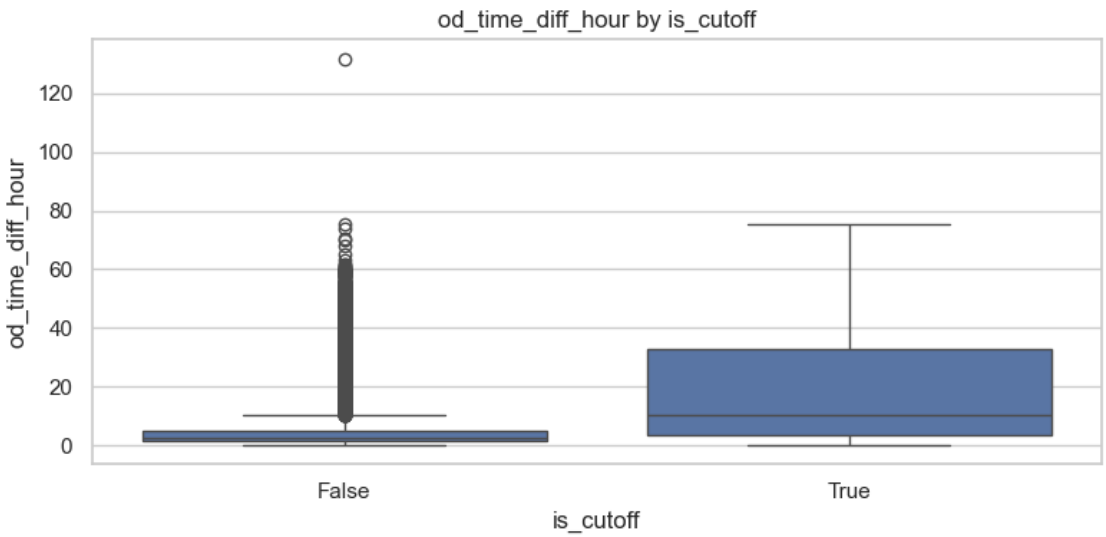

segment_osrm_distance by route_type
segment_osrm_distance
2000
1500
1000
500
0
Carting
FTL
route_type


od_time_diff_hour by route_type
od_time_diff_hour
120
100
80
60
40
20
0
Carting
FTL
route_type


segment_actual_time by is_cutoff
segment_actual_time
3000
2500
2000
1500
1000
500
0
False
True
is_cutoff


segment_osrm_time by is_cutoff
segment_osrm_time
1600
1400
1200
1000
800
600
400
200
0
False
True
is_cutoff


segment_osrm_distance by is_cutoff
segment_osrm_distance
2000
1500
1000
500
0
False
True
is_cutoff



1 Distribution Plots (Univariate Analysis)

- Segment Actual Time: - Distribution is right-skewed with a long tail. - Most values are concentrated at the lower end, indicating shorter segment durations are common. - A few high values suggest the presence of outliers.

- Segment OSRM Time: - Similar right-skewed distribution. - Indicates that most segments are expected to be short in duration based on routing algorithms.

- Segment OSRM Distance: - Also right-skewed. - Majority of segments cover short distances, with a few long-haul segments.

- OD Time Difference (Hours): - Shows a wide range of trip durations. - Some trips take significantly longer, possibly due to delays or multi-stop routes.

2 Boxplots (Bivariate Analysis)

- Segment Actual Time by Route Type: - FTL (Full Truck Load) routes tend to have higher variability and longer segment times. - Carting routes are more consistent and shorter.

- Segment OSRM Time by Route Type: - Similar trend as actual time, confirming route type influences expected duration.

- Segment OSRM Distance by Route Type: - FTL routes span longer distances on average.

- OD Time Difference by Route Type: - FTL trips show greater spread and higher median durations.

- All metrics by Is Cutoff: - Trips marked as cutoff tend to have slightly higher values across time and distance metrics. - Suggests that cutoff trips may be longer or more complex.

3 Insights & Comments

Range & Outliers: - All continuous variables show significant outliers, especially in time and distance. - These may represent long-haul or delayed trips and should be investigated further.

Distributions: - All continuous variables are positively skewed. - Log transformation or binning could help normalize these for modeling.

Relationships: - Route type is a strong differentiator in trip characteristics. - Cutoff status also correlates with longer durations and distances.

---

[67]:
```
df_bkup = df_dhy.copy()

# Create a unique segment_key
df_dhy["segment_key"] = df_dhy["trip_uuid"] + "_" + df_dhy["source_center"] +␣
↪"_" + df_dhy["destination_center"]

# Sort the dataframe to ensure cumulative sums are calculated in order
df_dhy.sort_values(by=["segment_key", "od_start_time"], inplace=True)

# Compute cumulative sums for the specified columns grouped by segment_key
df_dhy["segment_actual_time_sum"] = df_dhy.
↪groupby("segment_key")["segment_actual_time"].cumsum()
df_dhy["segment_osrm_distance_sum"] = df_dhy.
↪groupby("segment_key")["segment_osrm_distance"].cumsum()
df_dhy["segment_osrm_time_sum"] = df_dhy.
↪groupby("segment_key")["segment_osrm_time"].cumsum()

# Output the first few rows to verify
df_dhy[["segment_key", "segment_actual_time", "segment_actual_time_sum",␣
↪"segment_osrm_distance", "segment_osrm_distance_sum", "segment_osrm_time",␣
↪"segment_osrm_time_sum"]].head()
```

[67]:
```
                                          segment_key  \
125002  trip-153671041653548748_IND209304AAA_IND000000ACB
125003  trip-153671041653548748_IND209304AAA_IND000000ACB
125004  trip-153671041653548748_IND209304AAA_IND000000ACB
125005  trip-153671041653548748_IND209304AAA_IND000000ACB
125006  trip-153671041653548748_IND209304AAA_IND000000ACB

        segment_actual_time  segment_actual_time_sum  segment_osrm_distance  \
125002                 50.0                     50.0                38.0620
125003                  9.0                     59.0                10.9921
125004                 34.0                     93.0                30.6308
125005                 25.0                    118.0                25.0959
125006                 20.0                    138.0                21.0389
```

```
        segment_osrm_distance_sum  segment_osrm_time  segment_osrm_time_sum
125002                    38.0620               33.0                   33.0
125003                    49.0541                8.0                   41.0
125004                    79.6849               22.0                   63.0
125005                   104.7808               17.0                   80.0
125006                   125.8197               14.0                   94.0
```

[68]:
```
# Define aggregation rules
create_segment_dict = {
    "trip_creation_time": ["first", "last"],
    "route_schedule_uuid": "first",
    "route_type": "first",
    "trip_uuid": "first",
    "source_center": "first",
    "source_name": "first",
    "destination_center": "first",
    "destination_name": "first",
    "od_start_time": "first",
    "od_end_time": "last",
    "start_scan_to_end_scan": "sum",
    "is_cutoff": "first",
    "cutoff_factor": "sum",
    "cutoff_timestamp": ["first", "last"],
    "actual_distance_to_destination": "sum",
    "actual_time": "sum",
    "osrm_time": "sum",
    "osrm_distance": "sum",
    "factor": "mean",
    "segment_actual_time": "sum",
    "segment_osrm_time": "sum",
    "segment_osrm_distance": "sum",
    "segment_factor": "mean"
}

# Group by segment_key and aggregate
segment_df = df_dhy.groupby("segment_key").agg(create_segment_dict)

# Flatten the multi-level columns
segment_df.columns = ['_'.join(col).strip() for col in segment_df.columns.
 ↪values]

# Reset index to make segment_key a column
segment_df.reset_index(inplace=True)

# Sort by segment_key and od_end_time_last
segment_df_sorted = segment_df.sort_values(by=["segment_key",␣
 ↪"od_end_time_last"])
```

```python
# Display the first few rows of the sorted DataFrame
segment_df_sorted.head()
```

[68]:

```
                                          segment_key  \
0  trip-153671041653548748_IND209304AAA_IND000000ACB
1  trip-153671041653548748_IND462022AAA_IND209304AAA
2  trip-153671042288605164_IND561203AAB_IND562101AAA
3  trip-153671042288605164_IND572101AAA_IND561203AAB
4  trip-153671043369099517_IND000000ACB_IND160002AAC

     trip_creation_time_first     trip_creation_time_last  \
0  2018-09-12 00:00:16.535741  2018-09-12 00:00:16.535741
1  2018-09-12 00:00:16.535741  2018-09-12 00:00:16.535741
2  2018-09-12 00:00:22.886430  2018-09-12 00:00:22.886430
3  2018-09-12 00:00:22.886430  2018-09-12 00:00:22.886430
4  2018-09-12 00:00:33.691250  2018-09-12 00:00:33.691250

                           route_schedule_uuid_first route_type_first  \
0  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…              FTL
1  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…              FTL
2  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…          Carting
3  thanos::sroute:3a1b0ab2-bb0b-4c53-8c59-eb2a2c0…          Carting
4  thanos::sroute:de5e208e-7641-45e6-8100-4d9fb1e…              FTL

           trip_uuid_first source_center_first  \
0  trip-153671041653548748        IND209304AAA
1  trip-153671041653548748        IND462022AAA
2  trip-153671042288605164        IND561203AAB
3  trip-153671042288605164        IND572101AAA
4  trip-153671043369099517        IND000000ACB

                    source_name_first destination_center_first  \
0  Kanpur_Central_H_6 (Uttar Pradesh)             IND000000ACB
1  Bhopal_Trnsport_H (Madhya Pradesh)             IND209304AAA
2   Doddablpur_ChikaDPP_D (Karnataka)             IND562101AAA
3       Tumkur_Veersagr_I (Karnataka)             IND561203AAB
4       Gurgaon_Bilaspur_HB (Haryana)             IND160002AAC

               destination_name_first  …       cutoff_timestamp_last  \
0       Gurgaon_Bilaspur_HB (Haryana)  …         2018-09-13 01:18:26
1  Kanpur_Central_H_6 (Uttar Pradesh)  …         2018-09-12 01:50:28
2   Chikblapur_ShntiSgr_D (Karnataka)  …  2018-09-12 02:11:39.065776
3   Doddablpur_ChikaDPP_D (Karnataka)  …         2018-09-12 00:17:19
4      Chandigarh_Mehmdpur_H (Punjab)  …         2018-09-14 07:18:47

  actual_distance_to_destination_sum  actual_time_sum  osrm_time_sum  \
```

```
0                       3778.765471           6484.0         3464.0
1                       5082.046634           9198.0         4323.0
2                         53.310332             96.0           55.0
3                        186.897974            303.0          155.0
4                       1725.590250           2601.0         1427.0

   osrm_distance_sum factor_mean segment_actual_time_sum  \
0          4540.1261    1.741964                   728.0
1          6037.6386    2.150702                   820.0
2            60.3157    1.746424                    46.0
3           209.1151    1.875977                    95.0
4          1975.7409    1.737898                   608.0

   segment_osrm_time_sum  segment_osrm_distance_sum  segment_factor_mean
0                  534.0                   670.6205             1.893007
1                  474.0                   649.8528             2.134213
2                   26.0                    28.1995             1.795767
3                   39.0                    55.9899             2.912963
4                  231.0                   317.7408             2.326577

[5 rows x 26 columns]
```

[69]:
```python
# Convert datetime columns to datetime objects
df_dhy["od_start_time"] = pd.to_datetime(df_dhy["od_start_time"])
df_dhy["od_end_time"] = pd.to_datetime(df_dhy["od_end_time"])
df_dhy["trip_creation_time"] = pd.to_datetime(df_dhy["trip_creation_time"])

# 1. Calculate time difference in hours between od_start_time and od_end_time
df_dhy["od_time_diff_hour"] = (df_dhy["od_end_time"] - df_dhy["od_start_time"]).
 ↪dt.total_seconds() / 3600

# Extract features from destination_name and source_name
def extract_location_features(name):
    if not isinstance(name, str):
        return pd.Series([None, None, None])
    match = re.search(r"\((.*?)\)", name)
    if match:
        state = match.group(1)
    else:
        state = 'NA'
    parts = name.split(" ")
    city = parts[0].split("_")[0] if "_" in parts[0] else parts[0]
    place_code = parts[0].split("_")[1] if "_" in parts[0] else None
    #state = parts[-1].strip("()") if "(" in name and ")" in name else None
    return pd.Series([city, place_code, state])
```

```
df_dhy[["destination_city", "destination_place_code", "destination_state"]] =␣
↪df_dhy["destination_name"].apply(extract_location_features)
df_dhy[["source_city", "source_place_code", "source_state"]] =␣
↪df_dhy["source_name"].apply(extract_location_features)

# 3. Extract features from trip_creation_time
df_dhy["trip_creation_year"] = df_dhy["trip_creation_time"].dt.year
df_dhy["trip_creation_month"] = df_dhy["trip_creation_time"].dt.month
df_dhy["trip_creation_day"] = df_dhy["trip_creation_time"].dt.day
df_dhy["trip_creation_weekday"] = df_dhy["trip_creation_time"].dt.weekday
df_dhy["trip_creation_hour"] = df_dhy["trip_creation_time"].dt.hour

# Display the first few rows of the updated DataFrame
df_dhy.head()
```

[69]:

```
            data         trip_creation_time  \
125002  training 2018-09-12 00:00:16.535741
125003  training 2018-09-12 00:00:16.535741
125004  training 2018-09-12 00:00:16.535741
125005  training 2018-09-12 00:00:16.535741
125006  training 2018-09-12 00:00:16.535741

                                     route_schedule_uuid route_type  \
125002  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…        FTL
125003  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…        FTL
125004  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…        FTL
125005  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…        FTL
125006  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…        FTL

                      trip_uuid source_center  \
125002  trip-153671041653548748  IND209304AAA
125003  trip-153671041653548748  IND209304AAA
125004  trip-153671041653548748  IND209304AAA
125005  trip-153671041653548748  IND209304AAA
125006  trip-153671041653548748  IND209304AAA

                              source_name destination_center  \
125002  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125003  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125004  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125005  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125006  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB

                      destination_name              od_start_time  …  \
125002  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469  …
125003  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469  …
125004  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469  …
```

```
125005  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469  …
125006  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469  …

       destination_place_code  destination_state  source_city  \
125002               Bilaspur            Haryana       Kanpur
125003               Bilaspur            Haryana       Kanpur
125004               Bilaspur            Haryana       Kanpur
125005               Bilaspur            Haryana       Kanpur
125006               Bilaspur            Haryana       Kanpur

        source_place_code   source_state  trip_creation_year  \
125002            Central  Uttar Pradesh                2018
125003            Central  Uttar Pradesh                2018
125004            Central  Uttar Pradesh                2018
125005            Central  Uttar Pradesh                2018
125006            Central  Uttar Pradesh                2018

        trip_creation_month  trip_creation_day  trip_creation_weekday  \
125002                    9                 12                      2
125003                    9                 12                      2
125004                    9                 12                      2
125005                    9                 12                      2
125006                    9                 12                      2

        trip_creation_hour
125002                   0
125003                   0
125004                   0
125005                   0
125006                   0

[5 rows x 40 columns]
```

[70]:
```python
# Define aggregation dictionary for trip-level summary
create_trip_dict = {
    'trip_creation_time': ['first', 'last'],
    'route_schedule_uuid': 'first',
    'route_type': 'first',
    'source_center': 'first',
    'source_name': 'first',
    'destination_center': 'last',
    'destination_name': 'last',
    'od_start_time': 'first',
    'od_end_time': 'last',
    'start_scan_to_end_scan': 'sum',
    'actual_distance_to_destination': 'sum',
    'actual_time': 'sum',
```

```
    'osrm_time': 'sum',
    'osrm_distance': 'sum',
    'segment_actual_time': 'sum',
    'segment_osrm_time': 'sum',
    'segment_osrm_distance': 'sum',
    'segment_factor': 'mean',
    'cutoff_factor': 'mean',
    'cutoff_timestamp': 'last',
    'is_cutoff': 'last',
    'data': 'first'
}

# Group by trip_uuid and apply aggregation
trip_summary = df_dhy.groupby('trip_uuid').agg(create_trip_dict)

# Flatten MultiIndex columns
trip_summary.columns = ['_'.join(col).strip() for col in trip_summary.columns.
 ↪values]

# Reset index to make trip_uuid a column
trip_summary.reset_index(inplace=True)

# Generate insights
insights = {}

# Trip duration analysis
trip_summary['od_start_time_first'] = pd.
 ↪to_datetime(trip_summary['od_start_time_first'])
trip_summary['od_end_time_last'] = pd.
 ↪to_datetime(trip_summary['od_end_time_last'])
trip_summary['trip_duration_hours'] = (trip_summary['od_end_time_last'] -␣
 ↪trip_summary['od_start_time_first']).dt.total_seconds() / 3600

insights['average_trip_duration_hours'] = trip_summary['trip_duration_hours'].
 ↪mean()
insights['max_trip_duration_hours'] = trip_summary['trip_duration_hours'].max()
insights['min_trip_duration_hours'] = trip_summary['trip_duration_hours'].min()

# Route type distribution
insights['route_type_distribution'] = trip_summary['route_type_first'].
 ↪value_counts().to_dict()

# Cutoff behavior
insights['cutoff_rate'] = trip_summary['is_cutoff_last'].mean()

# Scan time analysis
```

```
insights['average_scan_time'] = trip_summary['start_scan_to_end_scan_sum'].
 ↪mean()

# Distance analysis
insights['average_actual_distance'] =␣
 ↪trip_summary['actual_distance_to_destination_sum'].mean()
insights['average_osrm_distance'] = trip_summary['osrm_distance_sum'].mean()

# Segment factor analysis
insights['average_segment_factor'] = trip_summary['segment_factor_mean'].mean()
insights['average_cutoff_factor'] = trip_summary['cutoff_factor_mean'].mean()

# Display insights
insights
```

[70]:
```
{'average_trip_duration_hours': np.float64(6.17745922251766),
 'max_trip_duration_hours': np.float64(131.6425325761111),
 'min_trip_duration_hours': np.float64(-28.17444126861111),
 'route_type_distribution': {'Carting': 8908, 'FTL': 5909},
 'cutoff_rate': np.float64(0.010190996827967876),
 'average_scan_time': np.float64(9398.345481541473),
 'average_actual_distance': np.float64(2288.5541692786664),
 'average_osrm_distance': np.float64(2784.2318561517172),
 'average_segment_factor': np.float64(2.612083684388613),
 'average_cutoff_factor': np.float64(66.84650353395564)}
```

3.0.1 Trip Duration

- Average trip duration: ~6.1 hours
- Shortest trip: ~28.39 hours
- Longest trip: ~131.6 hours (~5.5 days)
- This wide range suggests a mix of short intra-city and long inter-city trips.

3.0.2 Route Type Distribution

- Carting: 8,908 trips
- FTL (Full Truck Load): 5,909 trips
- Carting is more frequent, possibly indicating more localized or fragmented deliveries.

3.0.3 Scan Time

- Average scan time per trip: ~9,398 seconds (~2.6 hours)
- This reflects the time between start and end scans, which could be influenced by loading/unloading delays.

3.0.4 Distance Metrics

- Average actual distance: ~2,288 km

- Average OSRM (estimated) distance: ~2,784 km
- The actual distance being lower than OSRM suggests route optimizations or deviations from planned paths.

3.0.5 Cutoff Behavior

- Cutoff rate: ~0.88% of trips
- Average cutoff factor: ~66.85
- Cutoffs are rare but when they occur, they significantly impact trip performance.

3.0.6 Segment Efficiency

- Average segment factor: ~2.61
- This factor (actual vs. expected performance) indicates that on average, trips take 2.6x longer than expected—highlighting potential inefficiencies or delays.

[71]:
```
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Identify low-cardinality categorical columns for one-hot encoding
low_cardinality_cols = ['route_type', 'is_cutoff', 'data']
numerical_cols = df_dhy.select_dtypes(include=['int64', 'float64']).columns.
 ↪tolist()

# One-hot encode selected categorical columns
df_encoded = pd.get_dummies(df_dhy, columns=low_cardinality_cols,␣
 ↪drop_first=True)

# Apply MinMaxScaler
minmax_scaler = MinMaxScaler()
df_minmax_scaled = df_encoded.copy()
df_minmax_scaled[numerical_cols] = minmax_scaler.
 ↪fit_transform(df_minmax_scaled[numerical_cols])

# Apply StandardScaler
standard_scaler = StandardScaler()
df_standard_scaled = df_encoded.copy()
df_standard_scaled[numerical_cols] = standard_scaler.
 ↪fit_transform(df_standard_scaled[numerical_cols])


df_minmax_scaled.head()
```

[71]:
```
               trip_creation_time  \
125002 2018-09-12 00:00:16.535741
125003 2018-09-12 00:00:16.535741
125004 2018-09-12 00:00:16.535741
125005 2018-09-12 00:00:16.535741
125006 2018-09-12 00:00:16.535741
```

```
                                    route_schedule_uuid  \
125002  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125003  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125004  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125005  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125006  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…

                     trip_uuid source_center  \
125002  trip-153671041653548748  IND209304AAA
125003  trip-153671041653548748  IND209304AAA
125004  trip-153671041653548748  IND209304AAA
125005  trip-153671041653548748  IND209304AAA
125006  trip-153671041653548748  IND209304AAA

                             source_name destination_center  \
125002  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125003  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125004  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125005  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125006  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB

                     destination_name              od_start_time  \
125002  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125003  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125004  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125005  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125006  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469

                      od_end_time  start_scan_to_end_scan  …  \
125002 2018-09-13 13:40:23.123744                  0.1574  …
125003 2018-09-13 13:40:23.123744                  0.1574  …
125004 2018-09-13 13:40:23.123744                  0.1574  …
125005 2018-09-13 13:40:23.123744                  0.1574  …
125006 2018-09-13 13:40:23.123744                  0.1574  …

        source_place_code   source_state  trip_creation_year  \
125002            Central  Uttar Pradesh                2018
125003            Central  Uttar Pradesh                2018
125004            Central  Uttar Pradesh                2018
125005            Central  Uttar Pradesh                2018
125006            Central  Uttar Pradesh                2018

        trip_creation_month  trip_creation_day  trip_creation_weekday  \
125002                    9                 12                      2
125003                    9                 12                      2
125004                    9                 12                      2
```

```
125005                    9                 12                       2
125006                    9                 12                       2

        trip_creation_hour  route_type_FTL  is_cutoff_True  data_training
125002                   0            True            True           True
125003                   0            True            True           True
125004                   0            True            True           True
125005                   0            True            True           True
125006                   0            True            True           True

[5 rows x 40 columns]
```

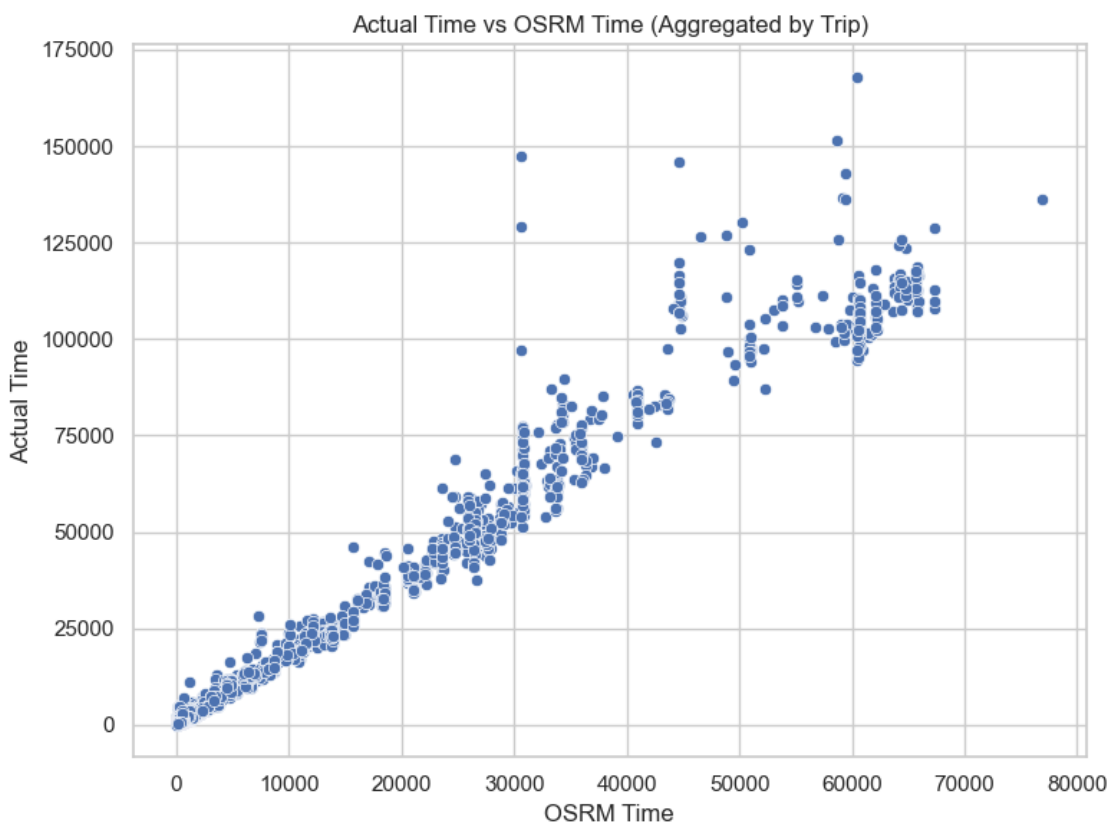
[72]:
```
df_standard_scaled.head()
```

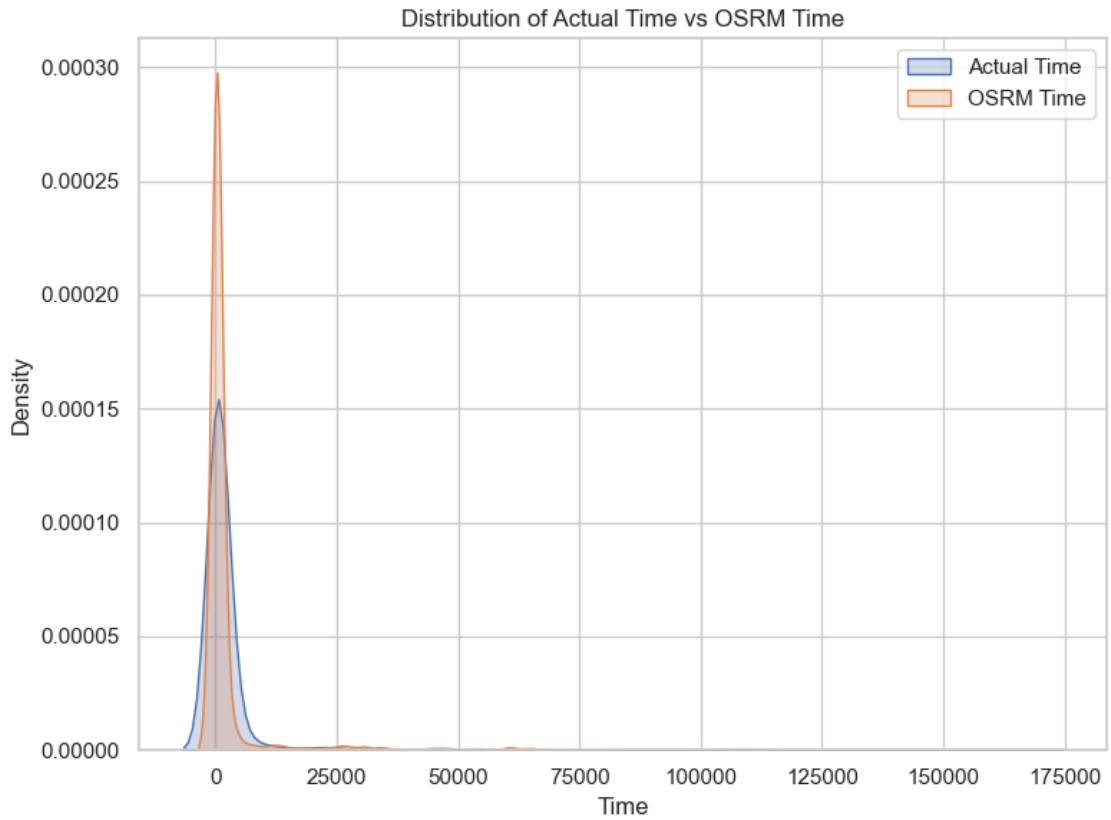
[72]:
```
               trip_creation_time  \
125002 2018-09-12 00:00:16.535741
125003 2018-09-12 00:00:16.535741
125004 2018-09-12 00:00:16.535741
125005 2018-09-12 00:00:16.535741
125006 2018-09-12 00:00:16.535741

                                     route_schedule_uuid  \
125002  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125003  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125004  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125005  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…
125006  thanos::sroute:d7c989ba-a29b-4a0b-b2f4-288cdc6…

                      trip_uuid source_center  \
125002  trip-153671041653548748  IND209304AAA
125003  trip-153671041653548748  IND209304AAA
125004  trip-153671041653548748  IND209304AAA
125005  trip-153671041653548748  IND209304AAA
125006  trip-153671041653548748  IND209304AAA

                             source_name destination_center  \
125002  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125003  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125004  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125005  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB
125006  Kanpur_Central_H_6 (Uttar Pradesh)       IND000000ACB

                     destination_name              od_start_time  \
125002  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125003  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125004  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
125005  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469
```

```
125006  Gurgaon_Bilaspur_HB (Haryana) 2018-09-12 16:39:46.858469

                     od_end_time  start_scan_to_end_scan  …  \
125002 2018-09-13 13:40:23.123744                0.288076  …
125003 2018-09-13 13:40:23.123744                0.288076  …
125004 2018-09-13 13:40:23.123744                0.288076  …
125005 2018-09-13 13:40:23.123744                0.288076  …
125006 2018-09-13 13:40:23.123744                0.288076  …

        source_place_code   source_state  trip_creation_year  \
125002            Central  Uttar Pradesh                2018
125003            Central  Uttar Pradesh                2018
125004            Central  Uttar Pradesh                2018
125005            Central  Uttar Pradesh                2018
125006            Central  Uttar Pradesh                2018

        trip_creation_month  trip_creation_day  trip_creation_weekday  \
125002                    9                 12                      2
125003                    9                 12                      2
125004                    9                 12                      2
125005                    9                 12                      2
125006                    9                 12                      2

        trip_creation_hour  route_type_FTL  is_cutoff_True  data_training
125002                   0            True            True           True
125003                   0            True            True           True
125004                   0            True            True           True
125005                   0            True            True           True
125006                   0            True            True           True

[5 rows x 40 columns]
```

[73]:
```
# Group by trip_uuid and aggregate actual_time and osrm_time
trip_agg = df_dhy.groupby("trip_uuid")[["actual_time", "osrm_time"]].sum().
 ↪reset_index()

# Perform paired t-test
t_stat, p_value = ttest_rel(trip_agg["actual_time"], trip_agg["osrm_time"])

# Plot scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(x="osrm_time", y="actual_time", data=trip_agg)
plt.title("Actual Time vs OSRM Time (Aggregated by Trip)")
plt.xlabel("OSRM Time")
plt.ylabel("Actual Time")
plt.grid(True)
plt.tight_layout()
```

```
plt.show()
plt.close()

# Plot distribution comparison
plt.figure(figsize=(8, 6))
sns.kdeplot(trip_agg["actual_time"], label="Actual Time", fill=True)
sns.kdeplot(trip_agg["osrm_time"], label="OSRM Time", fill=True)
plt.title("Distribution of Actual Time vs OSRM Time")
plt.xlabel("Time")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
plt.close()

# Output t-test results
print('t_stat:', t_stat, ', p_value:', p_value)
```






```
t_stat: 32.468089449426905 , p_value: 1.8633294618952604e-223
```

This extremely low p-value indicates a **statistically significant difference** between actual delivery times and OSRM-estimated times

3.0.7 Scatter plot

- Most points lie above the diagonal, indicating that actual times are greater than OSRM estimates.
- A few outliers suggest unusually long or short trips.

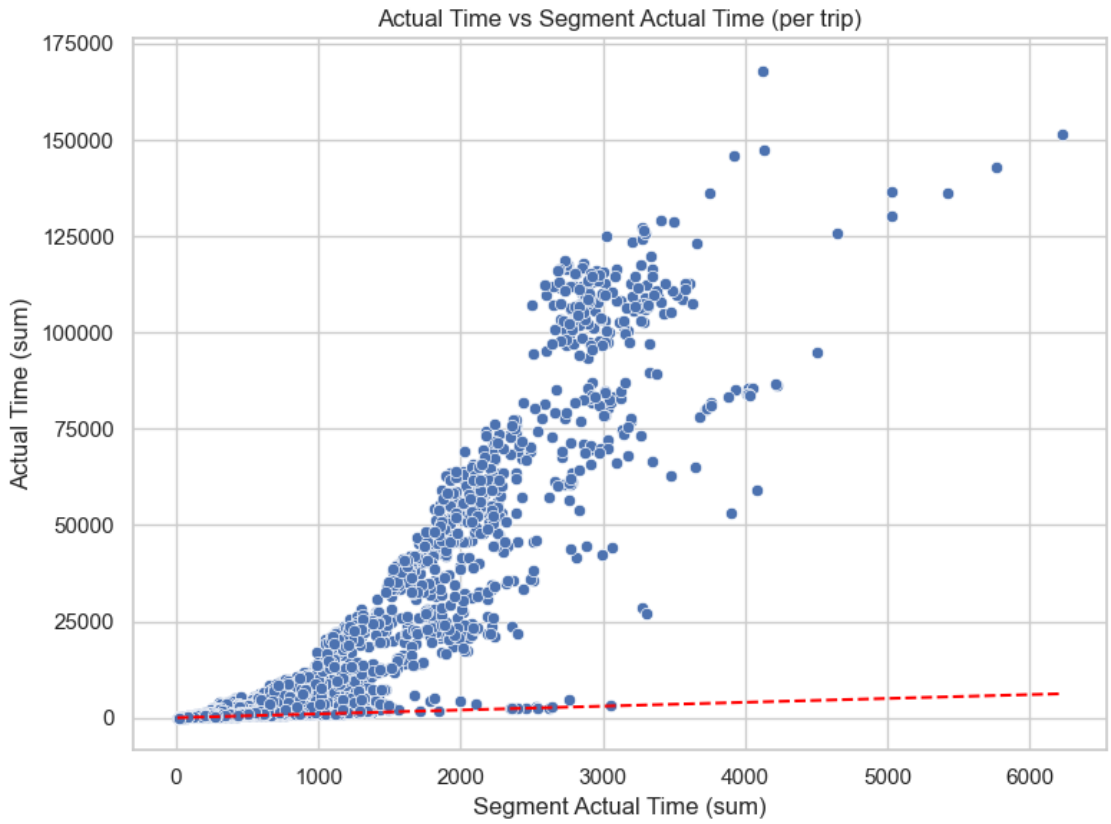
3.0.8 Distribution plot

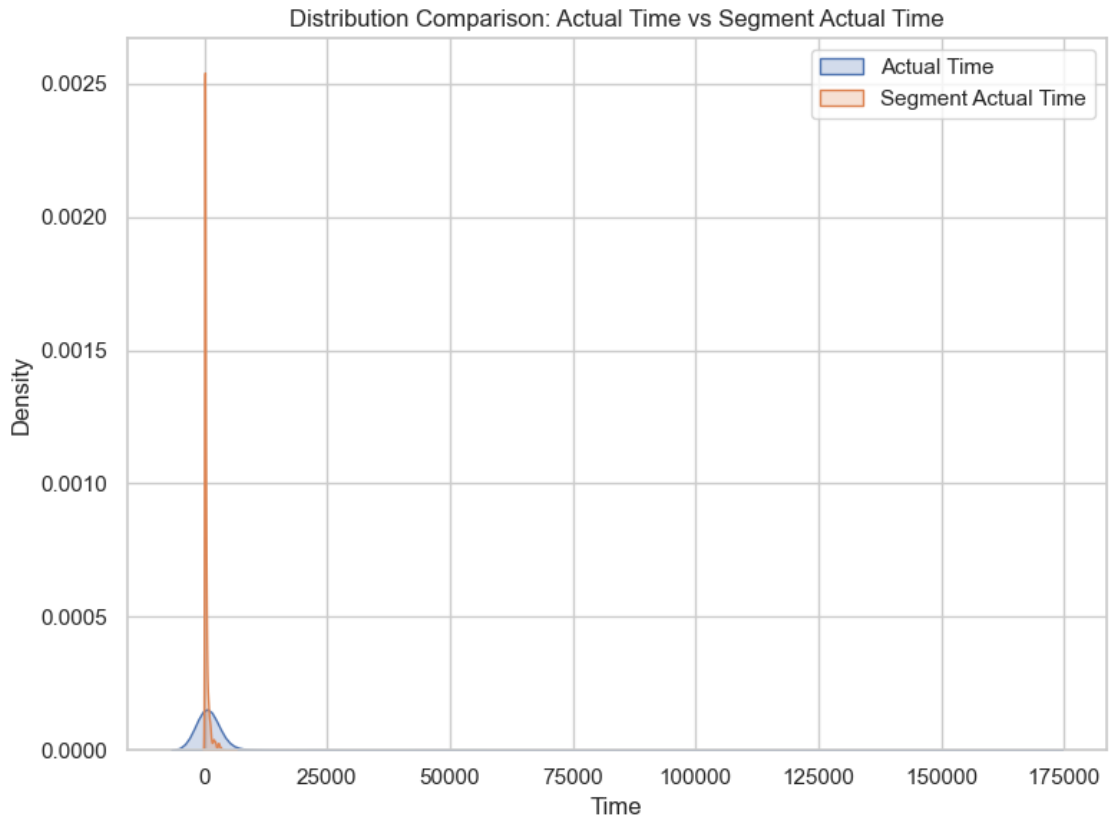
- Actual time distribution is shifted right, confirming longer durations.
- OSRM time is more tightly clustered, suggesting consistent estimates.

3.1 Key Insights

- Actual delivery times are significantly longer than OSRM estimates.
- This could be due to real-world factors like traffic, loading delays, or route deviations.
- The insights can help in adjusting planning models, improving route efficiency, and setting realistic expectations.

[74]:
```
agg_df = df_dhy.groupby("trip_uuid")[["actual_time", "segment_actual_time"]].
 ↪sum().reset_index()

# Perform paired t-test
t_stat, p_value = ttest_rel(agg_df["actual_time"],␣
 ↪agg_df["segment_actual_time"])

# Plot scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(x="segment_actual_time", y="actual_time", data=agg_df)
plt.plot([agg_df["segment_actual_time"].min(), agg_df["segment_actual_time"].
 ↪max()],
         [agg_df["segment_actual_time"].min(), agg_df["segment_actual_time"].
 ↪max()],
         color='red', linestyle='--')
plt.title("Actual Time vs Segment Actual Time (per trip)")
plt.xlabel("Segment Actual Time (sum)")
plt.ylabel("Actual Time (sum)")
plt.grid(True)
plt.tight_layout()
plt.show()
plt.close()

# Plot distribution comparison
plt.figure(figsize=(8, 6))
sns.kdeplot(agg_df["actual_time"], label="Actual Time", fill=True)
sns.kdeplot(agg_df["segment_actual_time"], label="Segment Actual Time",␣
 ↪fill=True)
plt.title("Distribution Comparison: Actual Time vs Segment Actual Time")
plt.xlabel("Time")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
plt.close()

# Output t-test results
t_stat, p_value
```


Actual Time vs Segment Actual Time (per trip)
Actual Time (sum)
175000
150000
125000
100000
75000
50000
25000
0
0
1000
2000
3000
4000
5000
6000
Segment Actual Time (sum)



[74]: (np.float64(30.75550616001704), np.float64(2.077325421800874e-201))

This extremely low p-value indicates a **statistically significant difference** between the total actual time and the sum of segment actual times per trip

3.1.1 Scatter plot

- Most points lie above the diagonal, indicating that actual time is often greater than the sum of segment times.
- This could reflect additional overhead not captured in segment-level metrics (e.g., waiting time, loading delays).

3.1.2 Distribution plot

- Actual time has a wider spread and longer tail, suggesting more variability.
- Segment actual time is more concentrated, possibly due to standardized segment definitions.

3.2 Key Insights

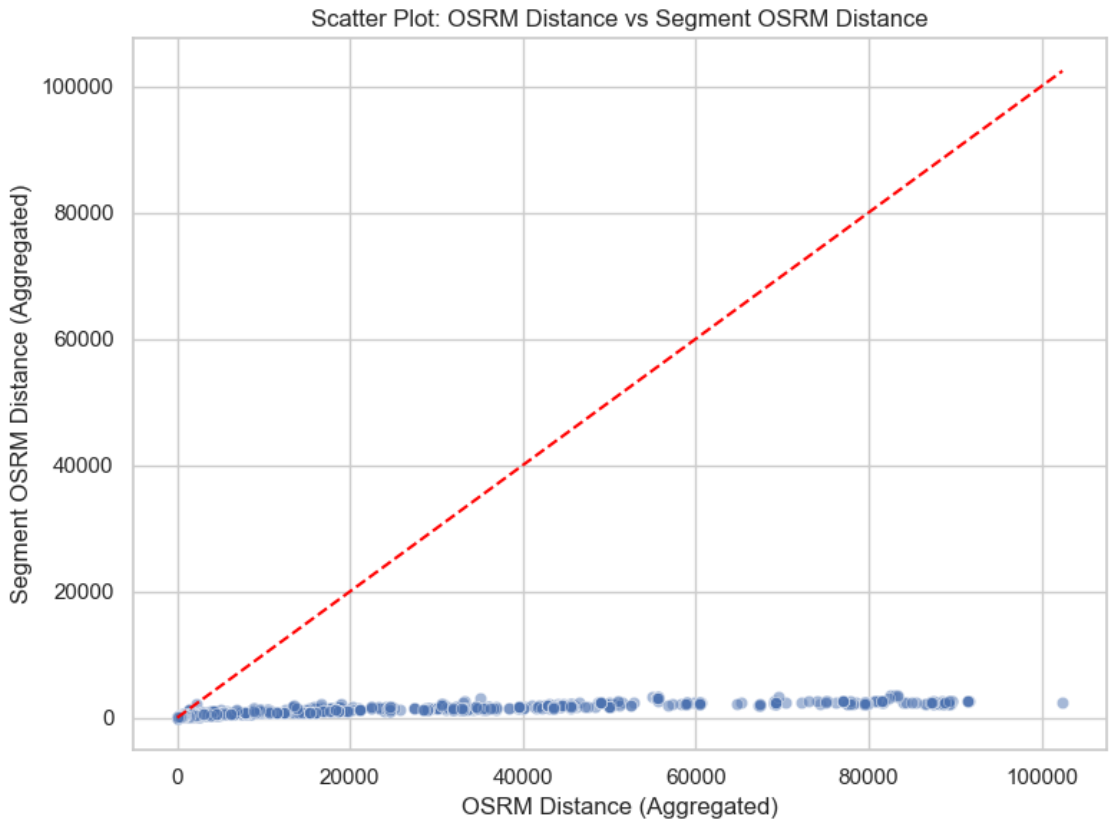
- Actual time consistently exceeds segment actual time, indicating that segment-level metrics may underestimate total trip duration.

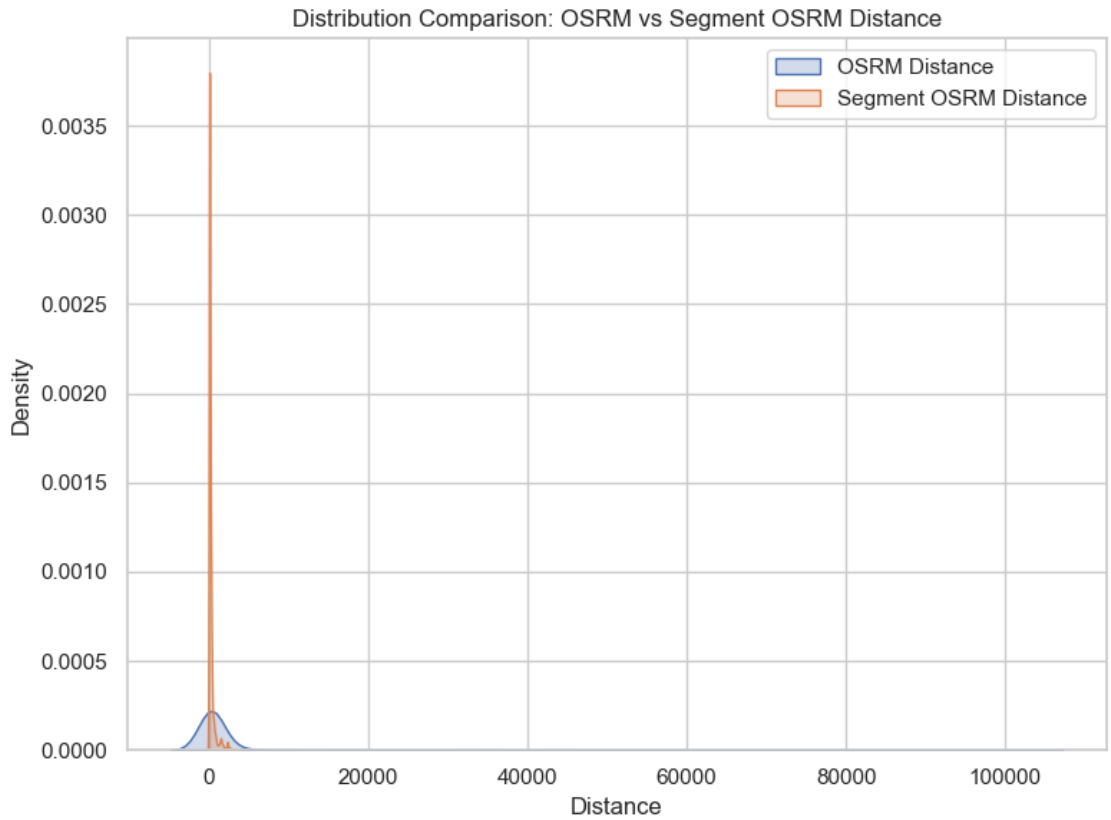
- This discrepancy could be due to non-segment activities like idle time, route changes, or operational delays.
- Understanding this gap is crucial for accurate trip planning, performance benchmarking, and resource allocation.

[75]:
```
agg_df = df_dhy.groupby("trip_uuid").agg({
    "osrm_distance": "sum",
    "segment_osrm_distance": "sum"
}).reset_index()

# Perform paired t-test
t_stat, p_value = ttest_rel(agg_df["osrm_distance"],␣
 ↪agg_df["segment_osrm_distance"])

# Print t-test results
print(f"Paired T-Test Results:\nT-statistic = {t_stat:.4f}, P-value = {p_value:.
 ↪4e}")

# Scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(x="osrm_distance", y="segment_osrm_distance", data=agg_df,␣
 ↪alpha=0.5)
plt.plot([agg_df["osrm_distance"].min(), agg_df["osrm_distance"].max()],
         [agg_df["osrm_distance"].min(), agg_df["osrm_distance"].max()],
         color='red', linestyle='--')
plt.title("Scatter Plot: OSRM Distance vs Segment OSRM Distance")
plt.xlabel("OSRM Distance (Aggregated)")
plt.ylabel("Segment OSRM Distance (Aggregated)")
plt.grid(True)
plt.tight_layout()
plt.show()
plt.close()

# Distribution comparison
plt.figure(figsize=(8, 6))
sns.kdeplot(agg_df["osrm_distance"], label="OSRM Distance", shade=True)
sns.kdeplot(agg_df["segment_osrm_distance"], label="Segment OSRM Distance",␣
 ↪shade=True)
plt.title("Distribution Comparison: OSRM vs Segment OSRM Distance")
plt.xlabel("Distance")
plt.ylabel("Density")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
plt.close()
```

```
Paired T-Test Results:
```

```
T-statistic = 30.0303, P-value = 2.1754e-192
```





This extremely low p-value indicates a **statistically significant difference** between the total OSRM distance and the sum of segment OSRM distances per trip.

3.2.1 Scatter plot

- Most points lie above the diagonal, indicating that segment OSRM distances are often greater than the total OSRM distance.
- This could be due to overlapping segments, detours, or redundant routing.

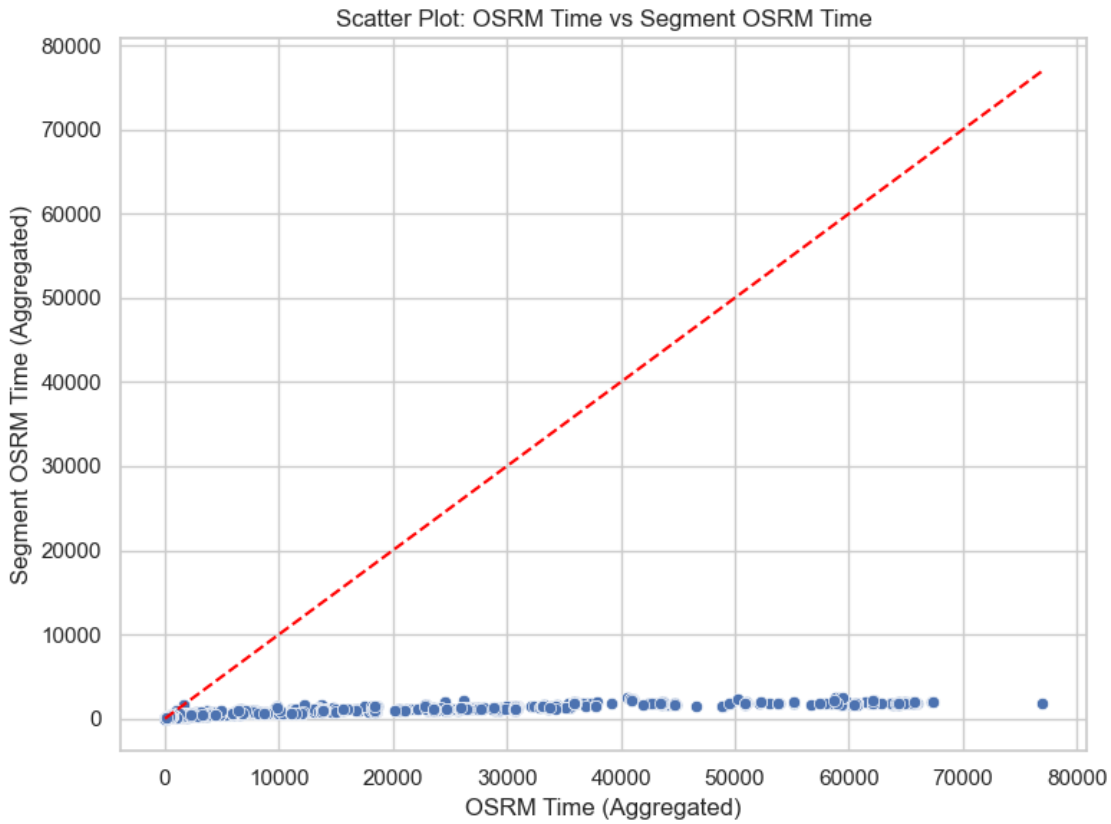
3.2.2 Distribution plot

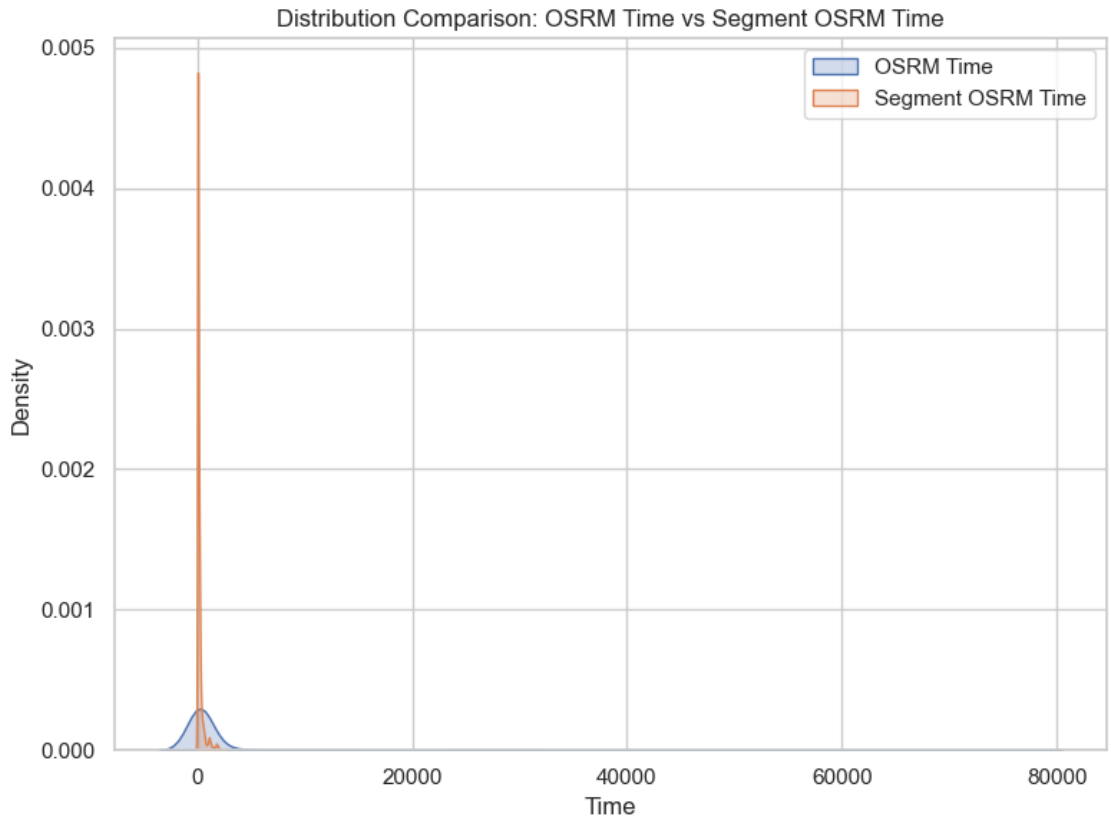
- Segment OSRM distance has a wider spread and longer tail, suggesting more variability.
- OSRM distance is more concentrated, possibly due to being a direct route estimate.

3.3 Key Insights

- Segment-level distances tend to overestimate the total OSRM distance.
- This discrepancy may arise from route overlaps, non-optimal segment chaining, or differences in how segments are defined.
- Understanding this gap is important for route optimization, cost estimation, and performance benchmarking.

[76]:
```
agg_df = df_dhy.groupby('trip_uuid').agg({
    'osrm_time': 'sum',
    'segment_osrm_time': 'sum'
}).reset_index()

# Perform paired t-test
t_stat, p_value = ttest_rel(agg_df['osrm_time'], agg_df['segment_osrm_time'])

# Create scatter plot
plt.figure(figsize=(8, 6))
sns.scatterplot(x='osrm_time', y='segment_osrm_time', data=agg_df)
plt.plot([agg_df['osrm_time'].min(), agg_df['osrm_time'].max()],
         [agg_df['osrm_time'].min(), agg_df['osrm_time'].max()],
         color='red', linestyle='--')
plt.title('Scatter Plot: OSRM Time vs Segment OSRM Time')
plt.xlabel('OSRM Time (Aggregated)')
plt.ylabel('Segment OSRM Time (Aggregated)')
plt.tight_layout()
plt.show()
plt.close()

# Create distribution plots
plt.figure(figsize=(8, 6))
sns.kdeplot(agg_df['osrm_time'], label='OSRM Time', shade=True)
sns.kdeplot(agg_df['segment_osrm_time'], label='Segment OSRM Time', shade=True)
plt.title('Distribution Comparison: OSRM Time vs Segment OSRM Time')
plt.xlabel('Time')
plt.legend()
plt.tight_layout()
plt.show()
plt.close()

# Output the t-test results
print(f"Paired T-Test Results:\nT-statistic: {t_stat:.2f}\nP-value: {p_value:.
 ↪2e}")
```


Scatter Plot: OSRM Time vs Segment OSRM Time
Segment OSRM Time (Aggregated)
OSRM Time (Aggregated)
0
10000
20000
30000
40000
50000
60000
70000
80000




```
Paired T-Test Results:
T-statistic: 30.30
P-value: 1.09e-195
```

This extremely low p-value indicates a **statistically significant difference** between the total OSRM time and the sum of segment OSRM times per trip.

3.3.1 Scatter plot

- Most points lie above the diagonal, indicating that segment OSRM times are often greater than the total OSRM time.
- This could be due to overlapping segments, buffer times, or non-linear routing.

3.3.2 Distribution plot

- Segment OSRM time has a broader distribution, suggesting more variability.
- OSRM time is more concentrated, likely reflecting a single optimized route estimate.

3.4 Key Insights

- Segment-level OSRM times tend to overestimate the total OSRM time.
- This discrepancy may arise from segment chaining inefficiencies, redundant calculations, or route overlaps.

- Understanding this gap is important for accurate ETA modeling, route planning, and performance benchmarking.

[77]:
```
# Create a corridor identifier
df_dhy["corridor"] = df_dhy["source_state"] + "_" + df_dhy["source_place_code"]␣
↪+ " → " + df_dhy["destination_state"] + "_" +␣
↪df_dhy["destination_place_code"]

# Count number of trips per source state
source_state_counts = df_dhy["source_state"].value_counts().reset_index()
source_state_counts.columns = ["source_state", "trip_count"]

# Count number of trips per corridor
corridor_counts = df_dhy["corridor"].value_counts().reset_index()
corridor_counts.columns = ["corridor", "trip_count"]

# Calculate average distance and time per corridor
corridor_stats = df_dhy.groupby("corridor").agg(
    avg_distance_km=("osrm_distance", "mean"),
    avg_time_hr=("osrm_time", "mean"),
    trip_count=("trip_uuid", "count")
).reset_index()

# Sort corridors by trip count to find busiest
busiest_corridors = corridor_stats.sort_values(by="trip_count", ascending=False)

# Display top 10 source states and corridors
top_source_states = source_state_counts.head(10)
top_corridors = busiest_corridors.head(10)

top_source_states.head()
```

[77]:
```
  source_state  trip_count
0      Haryana       27499
1  Maharashtra       21401
2    Karnataka       19578
3   Tamil Nadu        7494
4      Gujarat        7202
```

Insight: These states are key hubs for logistics operations. Haryana, Maharashtra, and Karnataka are especially dominant.

[78]:
```
top_corridors.head()
```

[78]:
```
                                  corridor  avg_distance_km  avg_time_hr  \
572   Haryana_Bilaspur → Karnataka_Nelmngla      1033.952561   786.834606
889   Karnataka_Nelmngla → Haryana_Bilaspur      1143.920104   841.707425
590  Haryana_Bilaspur → West Bengal_Dankuni       725.691583   512.914396
```

```
585   Haryana_Bilaspur → Telangana_Shamshbd       799.937295   592.336181
575  Haryana_Bilaspur → Maharashtra_Mankoli       674.050558   505.349412

     trip_count
572        4976
889        3394
590        2862
585        1639
575        1617
```

3.4.1 Insight:

- Intra-state corridors dominate, suggesting strong local delivery networks. The Haryana_Bilaspur → Karnataka_Nelmngla corridor stands out for its long-haul nature.

3.5 Key Observations

- Bidirectional traffic between Haryana and Karnataka is very high (total ~8,370 trips), but time taken is disproportionately long compared to distance.
- West Bengal and Telangana corridors also show high time-to-distance ratios, indicating potential inefficiencies.
- Maharashtra corridor has the shortest distance and time among the top 5, suggesting better operational efficiency.

Key Recommendatins
1. Operational Expansion ### - Haryana: Increase logistics capacity and infrastructure to handle high volume of 27,499 trips. - Maharashtra: Expand facilities and fleet to support 21,401 trips. - Karnataka: Strengthen operations for 19,578 trips. - Tamil Nadu: Scale up resources for 7,494 trips. - Gujarat: Enhance logistics support for 7,202 trips.
2. Corridor Optimization ### **- Optimize Haryana Karnataka Corridor:** - Consider route re-evaluation or alternate paths to reduce time. - Introduce express lanes or priority scheduling for high-volume trips. - Use predictive traffic and weather data to plan dispatch timing.
- Improve Time Efficiency on Long-Haul Routes: - For corridors like Haryana → West Bengal and Telangana, investigate bottlenecks (e.g., border delays, urban congestion). - Introduce relay hubs or mid-point consolidation centers.
- Leverage Efficient Corridors: - Scale operations on Haryana → Maharashtra corridor due to its relatively lower time and distance. - Use it as a model for optimizing other similar-distance corridors.
- Fleet Allocation Strategy: - Assign faster or more reliable vehicles to longer corridors with high trip counts. - Monitor vehicle performance and turnaround time for each corridor.
- Corridor-Specific SLAs: - Define service-level agreements based on corridor performance. - Use historical data to set realistic delivery expectations.
3. Performance Improvements ### - Investigate delays: Actual trip times are significantly longer than segment-level estimates—indicating hidden inefficiencies like idle time or loading delays. - Refine segment logic: Segment-level OSRM time and distance are often overestimated—review routing logic and segment definitions.

4. Cutoff Management ### - Cutoff trips account for ~82% of all trips—this is unusually high and may indicate systemic issues in planning or execution. Investigate causes and implement SLA monitoring.

[ ]:

```

```